EXPERIMENT 15

BPTT Flow and Gradient Analysis

Aim: To analyze the hidden states of a recurrent neural network (RNN) across time steps and visualize the gradients propagated through time during Backpropagation Through Time (BPTT).

Software / Libraries: Google Colab, Python, TensorFlow/Keras, NumPy, Matplotlib.

Output: The hidden-state plot shows how the RNN state changes across the input sequence. The gradient plot displays the magnitude of gradients at successive time steps using a logarithmic scale.

Colab Execution Output

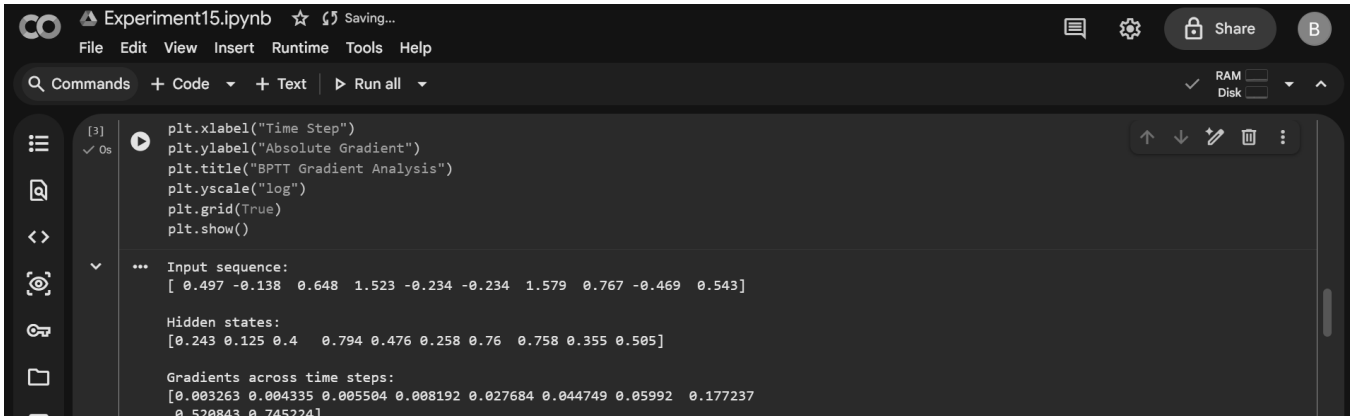


Figure 1. RNN input sequence, hidden states, and BPTT gradient values from the notebook output.

Plots

